Phase 1, Lesson 1

What Is Programming, and What Is Java?

Welcome! You don't need any background for this. We'll go one small step at a time.

1. What is programming?

Think of a **recipe**. A recipe is a list of exact steps:

1. Crack 2 eggs into a bowl.
2. Whisk for 30 seconds.
3. Pour into a hot pan.

A computer is like a very fast cook that has **no common sense at all**. It follows instructions exactly, in order, and it never guesses what you meant.

Programming means writing those instructions for a computer. The written instructions are called **code**, and a finished set of them is called a **program**.

2. What is Java?

Computers only understand electrical signals, which are just 1s and 0s. Humans can't comfortably write in 1s and 0s, so we use **programming languages**. These are languages that both humans and computers can work with.

Java is one of those languages. It's like choosing English or Spanish to write your recipe in.

3. Why Java is special

Java's big promise is **"Write once, run anywhere."**

Imagine you write your recipe once, and a **universal translator** in every kitchen reads it out in that kitchen's own language. That translator is called the **JVM (Java Virtual Machine)**. Because of it, the same Java program runs on Windows, Mac, Linux, and huge servers.

That's why banks, online stores, and apps used by millions of customers run their **backend** on Java. The backend is the hidden "kitchen" behind an app, where orders get processed and data gets stored. By the end of this journey, you'll be building those kitchens yourself.

4. A tiny peek (no need to understand it yet!)

```
void main() {
    System.out.println("Hello, future Java engineer!");
}
```

This program tells the computer one thing: *"Show this message on the screen."* That's it! This short modern style works in recent versions of Java (Java 25 and newer), so you skip a lot of the old, extra code beginners used to have to type.

Key words to remember

Word	Simple meaning
Program	A recipe of exact steps for the computer
Code	The written instructions
Programming language	A language humans and computers both understand
JVM	The universal translator that runs Java anywhere
Backend	The hidden kitchen behind an app

Your check-in question

Answer in your own words. It's fine to be casual:

1. **What is a program?** (Hint: think about the recipe.)
2. **Bonus:** What job does the JVM do?

There are no wrong answers here. Reply whenever you're ready, and then we'll move on to **Lesson 2: Setting up Java and running your very first program!**